



Errors & Uncertainties

1. Ruler



A distance of 15.4 cm is measured using a ruler with a graduation of 1 mm. Calculate the relative uncertainty.

2. Electron Mass



The mass of an electron is to be measured with an uncertainty of at most 2 %. Calculate the allowed absolute uncertainty.

3. Relative Error



A quantity a is measured with an uncertainty of 1 %. Determine the relative uncertainties of the quantities a^2 and $\sqrt{a}$.

4. Square



The side of a square is measured to have a length 12.35 mm with an uncertainty 0.05 mm. Calculate the area with its absolute uncertainty.

5. Density



Calculate the density of a metal cube with edge length (3.2 ± 0.1) cm and mass (88.21 ± 0.35) g. What material is it made of?

6. Velocity



A toy car covers a distance (3.29 ± 0.04) m in (1.08 ± 0.03) s. Calculate the velocity with its absolute uncertainty.

7. Sphere



A spherical object has a volume of (783.0 ± 1.8) cm³. Calculate the radius and its absolute uncertainty.

8. Maximum Uncertainty



Using the maximum uncertainty method, determine the uncertainty of $y = \cos(\omega t)$ for $t = (4.32 \pm 0.02)$ s and $\omega = (7.18 \pm 0.01)$ rad/s.